MILITARY SPECIFICATION

2,2' -METHYLENEBIS (4-METHYL-6-TERT-BUTYLPHENOL)

This Specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and quality assurance provisions for the manufacture and acceptance of one type of 2,2'-methylenebis (4-methyl-6-tert-butylphenol) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 6810
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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-286	- Propellants, Solid: Sampling, Examination and Testing
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1218	- ACS Chemicals

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific procurement functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovenment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D2905	- Statement on Number of Specimens Required to Determine the Average Quality of a Textile Material
ASTM D3951	- Standard Practice for Commercial Packaging
ASTM E168	- General Technique for Infrared Quantitative Analysis
ASTM E300-73	- Recommended practice for Sampling Industrial Chemicals

(Application for copies should be addressed to the American Society For Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standars), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Properties. The properties of the 2,2'-Methylenebis (4-methyl-6-tert-butylphenol) shall comply with the requirements given in Table I when determined in accordance with the applicable test paragraph.

TABLE I. Requirements

Characteristic	Requirement Min	Requirement Max	Applicable Test Method
Melting Range, °C	120	130	4.5.1
Particle Size, Less than 149 micron, %	99.5	-	4.5.2
Moisture, %	-	0.5	4.5.3
Ash, %	-	0.25	4.5.4

3.2 Infrared spectrum (IR). The IR spectrum shall contain the major peaks shown in Figure 1 when in accordance with 4.5.5.

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.3).

3.4 Workmanship. The manufacturer shall implement procedures and controls to assure that the process and the product produced are not compromised by foreign materials and contaminants or any other conditions which may degrade the composition. Determination of foreign materials shall be in accordance with 4.5.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual nor does it commit the Government to acceptance of any defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3)

b. Quality Conformance Inspection (see 4.4)

4.3 First article inspection.

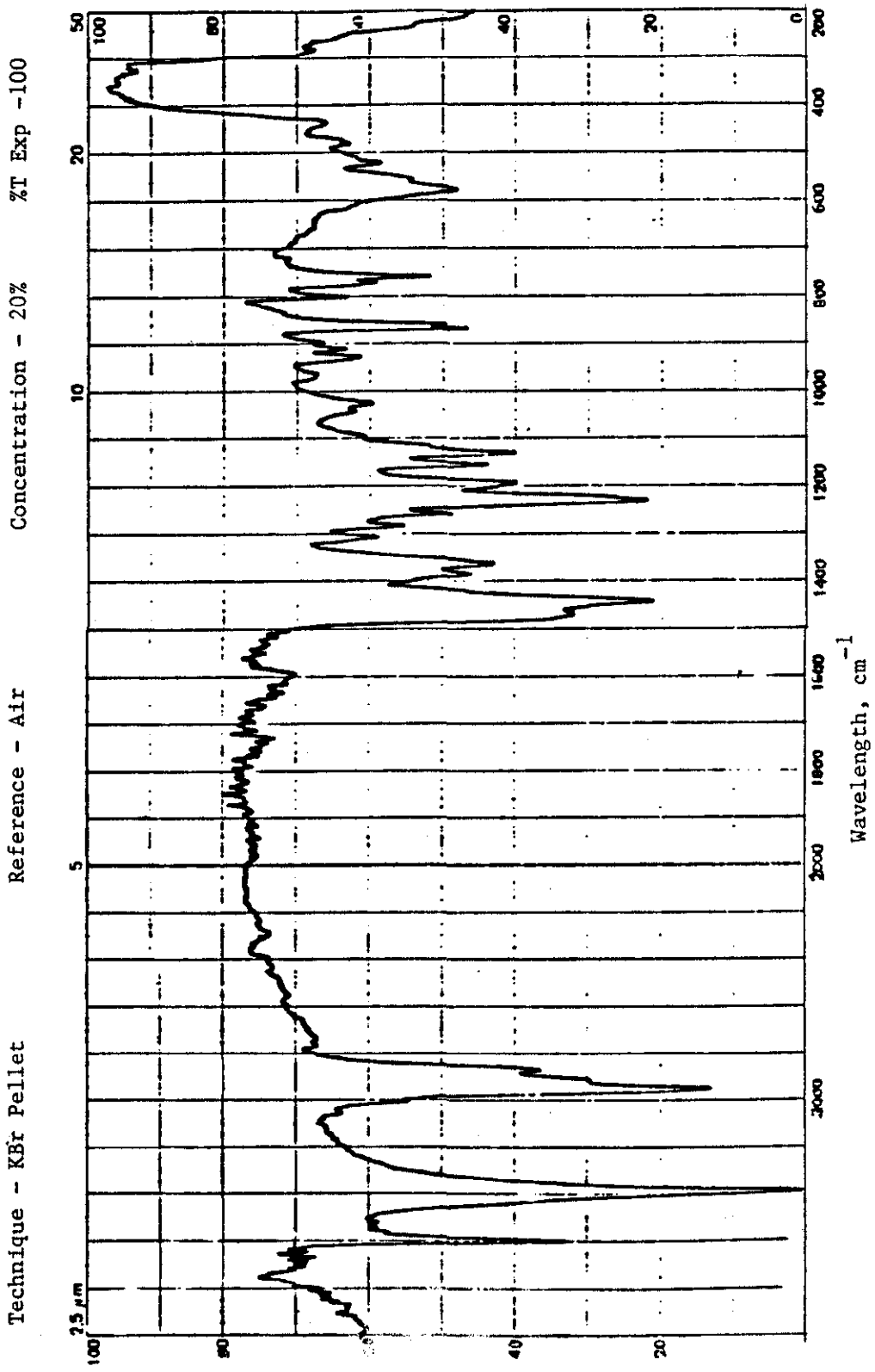
4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall be ten (10) pounds of 2,2'-Methylenebis (4-Methyl-6-tert-butylphenol). The sample shall be obtained from the first production lot which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspection to be performed. The sample will be subjected by the Government to any or all of the examinations or tests specified in Table II (see MIL-A-48078).

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Figure 1 - IR Spectrum for 2,2'-Methylenebis(4-Methyl-6-tert-butylphenol)

MAJOR PEAKS, cm^{-1}	FUNCTIONAL GROUP
3600, 3400	OH
2960,2920,2800	CH_3, CH_2
1440	Ring Structure
1390, 1370	Tertiary Butyl
1235	Alkyl phenol
870	H on Ring



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4.3.3 Rejection. See MIL-A-48078.

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TABLE II. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF	DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH / PARAGRAPH REFERENCE /INSPECTION METHOD
	Melting Range			3.1 / 4.5.1
	Particle Size			3.1 / 4.5.2
	Moisture			3.1 / 4.5.3
	Ash			3.1 / 4.5.4
	Infrared Spectrum			3.2 / 4.5.5
	Workmanship			3.4 / 4.5.6

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Lot formation. A lot shall consist of one or more batches of 2,2'-Methylenebis (4-methyl-6-tert-butylphenol) produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of 2,2'-Methylenebis (4-methyl-6-tert-butylphenol) that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous. The lot shall comply with the provisions for submission of product as specified in MIL-STD-105. The criteria and procedure for the assignment of lot numbers shall be in accordance with MIL-STD-1168. Also, MIL-A-48078 applies.

4.4.2 Examination. See MIL-A-48078.

a. Workmanship. Examination for workmanship shall be conducted on each sample selected for testing in accordance with 4.4.3.1. If any sample fails to meet any test requirement the batch represented by the sample shall be rejected.

4.4.3 Testing. PRECAUTION: This specification covers sampling and testing of toxic and hazardous material. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing the 2,2'-Methylenebis (4-Methyl-6-tert-butylphenol).

4.4.3.1 Sampling. Approximately 500 grams of the composition shall be selected from each batch to be samples using ASTM Method E300-73 for solids. Samples shall be selected for inspection in accordance with table below. If any sample fails to meet any test requirement the batch represented by the sample shall be rejected. Each required test shall be performed in duplicate. No composite samples shall be used. The classification of defects shall be as given in Table III.

Number of containers in batch	Number of containers to be sampled
2 - 100	2
100 - 200	5
200 - 300	8
300 - 500	10
500 - 1000	13
1000 - 3000	20
3000 - 10,000	32

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TABLE III. Classification of defects

Test/Examination	Requirement Paragraph	Defect Classification
Melting Range	3.1	Major
Particle Size	3.1	Major
Moisture	3.1	Major
Ash	3.1	Major
Infrared Spectrum	3.2	Major
Workmanship	3.4	Major

4.4.4 Inspection equipment. For the performance of all tests and examinations specified in 4.4 and 4.5, commercial inspection equipment should be employed. The contractor shall have available, and utilize correctly, this equipment and is charged with the responsibility of assuring that proper calibration procedures are followed.

4.5 Methods of inspection. All tests given in this section shall be performed using prescribed procedures for replicate determination given in standard analytical chemistry textbooks or ASTM D2905. Also, unless otherwise specified herein, all chemicals shall be Reagent Grade or ACS Grade in accordance with MIL-STD-1218. See 6.3 for use of equivalent test methods.

4.5.1 Melting point range.

4.5.1.1 Apparatus.

a. Fisher-Johns melting point apparatus

b. Micro spatula

c. Micro cover glasses, 18mm diameter

4.5.1.2 Procedure.

a. Place a cover glass on the heating plate which is under the magnifier. Place a small amount of sample on top of the cover glass and place a second cover on top of the sample.

b. Turn the toggle switch on. Make sure that the temperature rheostat is turned completely to the left or counter-clockwise.

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c. The temperature rheostat is numbered from 1 to 100 in units of heat. By turning this in the clockwise direction slowly, a change in temperature is recorded on a thermometer. Observe the melting point by looking through the magnifying glass at the sample and observing the temperature on the thermometer. When the sample is seen to soften (begins to melt), note the temperature and record to the nearest 0.1°C. The increase in temperature should not be more than one degree per minute in order to ensure that the temperature recorded on the thermometer is the same as that of the sample.

d. Continue observing the sample until all solid has turned to liquid. Note the temperature ad record to the nearest 0.1°C. Both the first drop and complete melting temperatures must be within the required range.

4.5.2 Particle size.

a. Weigh 100 grams of sample into a beaker.

b. Add approximately 1 gram of wetting agent (see 6.5).

c. Add sufficient water to create a thin, uniform slurry.

d. Pour the thin slurry onto a No. 100 screen, and wash with tap water, brushing gently until it is apparent that no more material will pass through the screen.

e. Dry residue in a drying oven at 55° to 60°C for 2 hours minimum and weigh.

f. Determine particle size distribution as follows:

$$\% \text{ Through No. 100 screen} = \frac{(W - R)}{W} \frac{100}{}$$

Where:

W = weight of sample
R = weight of residue

% Through No. 100 screen = % less than 149 Microns

4.5.3 Moisture. The moisture content shall be determined in accordance with Method 101.5 of MIL-STD-286.

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4.5.4 Ash. Dry the sample for 2 hr at 105°C and weigh 10 grams to the nearest 0.1mg. Burn directly over a flame in a 100 ml tared platinum crucible that has been heated to constant weight and weighed to the nearest 0.1mg. Add the sample in portions. The sample should burn gently and the portions should be added as the flame subsides. Continue heating with a burner only as long as the residue burns with a flame. Transfer the crucible to a muffle furnace and heat at 550 to 600°C for 3 hr. Allow crucible to cool and then transfer it, while still warm, to a desiccator. When the crucible has cooled to room temperature, weigh accurately to the nearest 0.1 mg.

$$\text{Ash, \%} = \frac{A \times 100}{B}$$

Where:

A = grams of ash
B = grams of sample used

4.5.5 Infrared spectrum. An infrared spectrum shall be obtained using a procedure which has had prior approval of the Contracting Officer. ASTM E168 shall be used as a basis of the procedure.

4.5.6 Workmanship. Visually examine the sample for contamination or foreign matter.

5. PACKAGING

5.1 Packaging.

5.1.1 Level A. - When applicable packaging shall be as specified in the contract.

5.1.2 Level B. - When applicable packaging shall be as specified in the contract.

5.1.3 Level C. - Unless otherwise specified in the contract or order, packaging and packing shall be in accordance with ASTM-D-3951.

5.2. Marking. - Unless otherwise specified in contract or order, marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The 2,2'-Methylenebis (4-methyl-6-tert-butylphenol) covered by this specification is intended for use in the M864 propellant.

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6.2 Ordering data. See MIL-A-48078.

6.3 Equivalent test methods. The test methods given in this specification are the official methods to be used. The contractor may request using other methods providing that the proposed method is equivalent (accuracy and precision) to the method given in this specification. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, ARDEC, ATTN: AMSMC-QAR-R, Picatinny Arsenal, NJ 07806-5000. This description should include, but not be limited to, the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required (see MIL-I-45208).

6.4 Subject term (key word) listing.

Propellant
Antioxidant
Projectile

6.5 Wetting agent. Rohm & Hass Triton X-25 or equivalent.

Custodian:
Army-AR

Preparing activity
Army-AR
(Project 6810-A007)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-70605	2,2' METHYLENEBIS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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